YUSUF ADEL ABOUZEID

Artificial Intelligence & Embedded Systems Engineer | NLP, LLM & Edge AI Specialist

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SUMMARY

High-impact AI & Embedded Systems Engineer specializing in Natural Language Processing (NLP), Large Language Models (LLMs), and Edge AI. 1st Place Winner at the National AI Hackathon for engineering EVA-AI, a zero-hallucination Clinical Decision Support RAG platform, and Top 6 National Finalist at RoboDam 2026 for building RoboApex, an autonomous ESP-NOW microgrid gateway with embedded neural cortex. Proven expertise in fine-tuning transformer models (DeBERTa-v3, AraBERT), hybrid search (Dense + BM25 + Cross-Encoder), distributed IoT mesh networks, and production RESTful APIs. NVIDIA DLI and ITI certified in Generative AI.

HONORS & AWARDS

Top 6 National Finalist — RoboDam 2026 National Hackathon | Damanhour University & ITI (Sep 2026)

- Handpicked by the committee among the **TOP 6 standout projects** across Egypt (out of 65 university teams) to demonstrate RoboApex live before H.E. the Minister of Higher Education & Scientific Research and university leadership.

1st Place Winner — National AI Hackathon | Orange Digital Center, CREATIVA, ITIDA, TIEC & INSTANT (Aug 2026)

- Awarded ■ **1st Place** among top competing teams for developing EVA-AI, an evidence-grounded clinical decision-support platform featuring hybrid search reranking, multi-tier query caching, and a 5-layer safety guardrail workflow.

TECHNICAL SKILLS

Programming & Embedded: Python (Proficient), C/C++ (ESP32 Embedded), SQL, TypeScript, Java (Familiar), HTML/CSS

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), LLMs (Claude Sonnet 4.6, Gemini, LLaMA 3.3 70B), Prompt Engineering, Grounded Generation, Structural Page-Level Citations, Multi-Turn Dialogue

NLP & Transformers: Hugging Face Transformers, AraBERT, DeBERTa-v3, BAAI/bge-small, Cross-Encoder Reranking (ms-marco-MiniLM), Fine-Tuning, Text Preprocessing, Sentiment Analysis

IoT & Edge Systems: ESP-NOW Mesh Protocol (Sub-3ms P2P), SCADA WebServer, Edge AI Neural Cortex, Dynamic Peak Shaving, ADC Sensor Interfacing, Solar Inverter ATS

Information Retrieval & Search: Hybrid Search, BM25 Lexical Retrieval, Reciprocal Rank Fusion (RRF), Vector Databases (ChromaDB), Cosine Similarity, Multi-Tier Caching (L1/L2/L3)

Computer Vision & ML: PyTorch, OpenCV, Object Detection (Bounding Boxes), CNNs, Model Regularization (Dropout, Weight Decay), Metrics (Precision@K, Hit-Rate)

Frameworks & DevOps: FastAPI, Flask (REST APIs), Next.js 15, React (Vite), Sentry Observability, Git, GitHub Actions (CI/CD), Docker, Linux

FEATURED ENGINEERING PROJECTS

RoboApex — Autonomous Smart Microgrid Gateway & Edge AI Energy Manager | Lead Embedded & AI Systems Engineer (Aug – Sep 2026)

- Engineered an autonomous, decentralized Microgrid Gateway and HEMS across distributed ESP32 nodes using the ESP-NOW peer-to-peer protocol for zero-latency (<3ms), offline-resilient communication without Wi-Fi router dependency.
- Implemented an Edge AI Neural Cortex directly on ESP32 with predictive thermal modeling (dT/dt) to forecast room temperature 30 mins ahead, PMV thermal comfort scoring, and real-time current anomaly detection.
- Engineered automated Dynamic Peak Shaving to protect solar inverters and household infrastructure by shedding high-power EV charging loads during grid stress with 150-unit hysteresis deadband.
- Developed embedded SCADA dashboard with Groq LLaMA 3.3 70B conversational AI for natural-language telemetry querying and automated device action callbacks.

EVA-AI — Zero-Hallucination Clinical Decision Support Platform | Lead AI Engineer & Architect (Aug 2026)

- Architected a zero-hallucination Clinical Decision Support RAG system over official medical guidelines (NICE NG243), achieving ≥ 80% Hit-Rate and ≥ 0.82 Precision@K on an 18-query benchmark suite.
- Engineered a Dual-Engine Hybrid Retrieval Pipeline fusing Dense Cosine ChromaDB vectors, Okapi BM25 lexical search via Reciprocal Rank Fusion (RRF k=60), and Cross-Encoder Transformer Reranking (ms-marco-MiniLM-L-6-v2).
- Developed a 5-Layer Safety & Guardrail Workflow featuring sub-3ms pre-retrieval drug/dosage refusal, domain boundary enforcement, and zero-network local embedding fallback (BAAI/bge-small-en-v1.5).
- Built full-stack production application with FastAPI 0.115 backend, Next.js 15 (TypeScript) monomorphic soft-UI frontend with English/Arabic RTL layout parity, and Sentry distributed tracing with automated PHI/PII sanitization.

Fact Forge — Automated Misinformation & Fake News Detection | Core Model Developer (May – Jun 2026)

- Built an end-to-end fake news detection pipeline achieving 95.6% classification accuracy on the WELFake dataset (72,000 news records).

- Fine-tuned a DeBERTa-v3 transformer model using custom weight decay and dropout regularization to prevent overfitting and improve generalization.
- Engineered RESTful Flask API endpoints integrated with a responsive React (Vite) dashboard for real-time news credibility scoring.

Mesdaq AI — Arabic Misinformation Classification System | Core Model Developer (Jan 2026)

- Fine-tuned AraBERT transformer model to accurately classify Arabic digital news articles and mitigate misinformation.
- Designed a specialized Arabic NLP preprocessing pipeline handling morphological complexity, diacritics, and normalization.

Freelance AI/Python Developer — Kafiil Platform | Digital Egypt Youth Program (Oct 2025 – Feb 2026)

- **Recommendation Systems:** Developed a collaborative filtering product-recommendation engine for an e-commerce platform.
- **Computer Vision:** Implemented an automated vehicle detection tool using OpenCV and Python with 24-hour turnaround.
- **Arabic Sentiment Analysis:** Built an Arabic customer-review sentiment classifier outputting label distributions and analytics.

EDUCATION

B.Sc. in Computer Science, Major in Artificial Intelligence | Future University in Egypt (FUE) (2023 – Present)

- Cumulative GPA: **3.38 / 4.00** | Expected Graduation: 2027 | Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Embedded Systems, Data Structures & Algorithms, Database Systems.

CERTIFICATIONS & LEADERSHIP

- NVIDIA DLI Certificate — Generative AI & Deep Learning (ITI - Information Technology Institute)
- Responsible & Trustworthy AI Certification (SAS Institute)
- AI for Business Specialist — NTI (National Telecommunication Institute) — Passed with 96% Score
- Co-organizer, GDG (Google Developer Group) Community — FUE Chapter (2024 – 2025): Organized technical workshops and hackathons on AI/ML for 200+ students.